



Connecting Neuroscience and Machine Learning

Don't get Eaten by the Tiger

Name: _____ Date: _____

Neuroscience of Decision Making

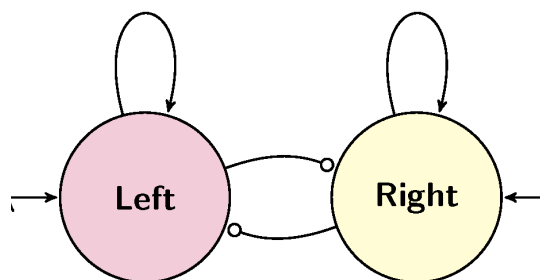
Exercise 1: How does a Neuron Work

What is the basic unit of information for a neuron?

How do we encode it mathematically?

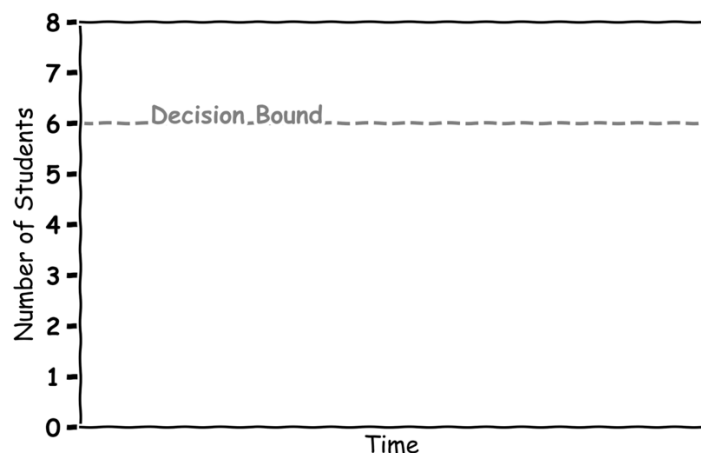
Exercise 2: Two Populations of Neurons

Label the graph:



Exercise 3: Two Populations of Neurons (Students)

Count and time the number of students response to over time sketch the left group in red and the right in green.





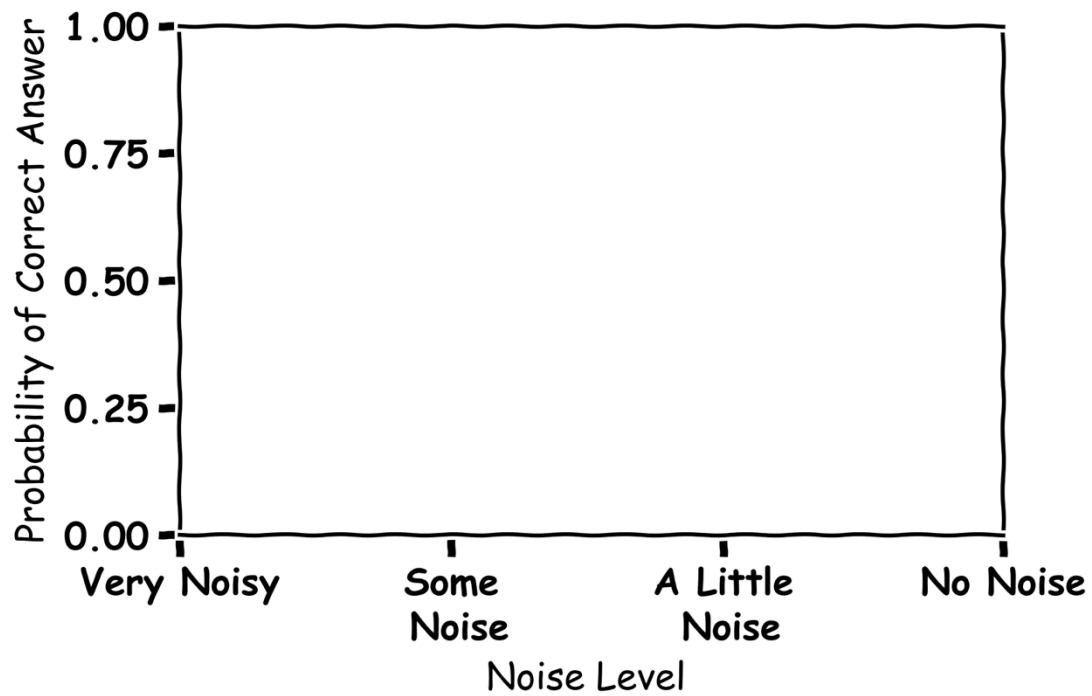
Making Decisions

Exercise 4: Are you Correct

Label the graph:

Where is the 50-50 chance line?

Fill in the graph with your answers:





Machine Learning of Decision Making



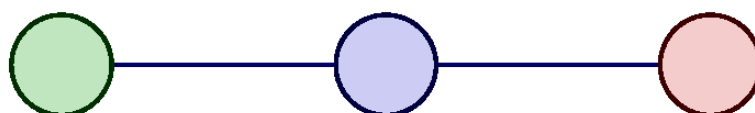
Exercise 5: The Machine Learning Algorithm

What is a Machine Learning algorithm?

Give an example of Machine Learning in your day to day life?

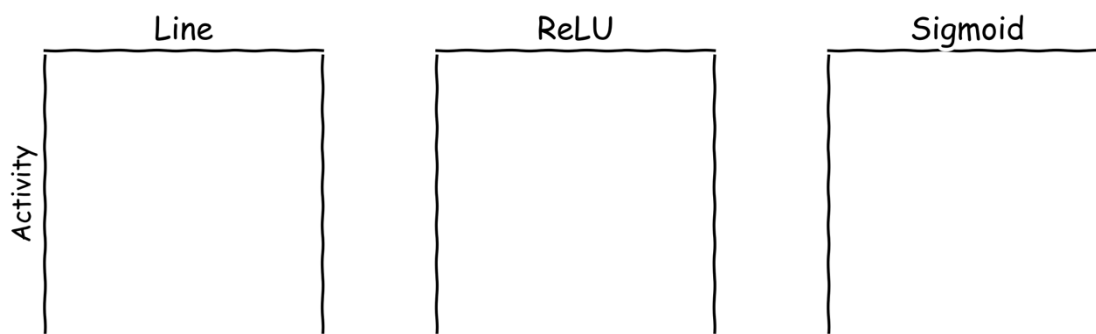
Exercise 6: The Node

Label the graph:



Exercise 7: Activation Functions

7.1 Draw the functions could be used in a node and what are they used for?

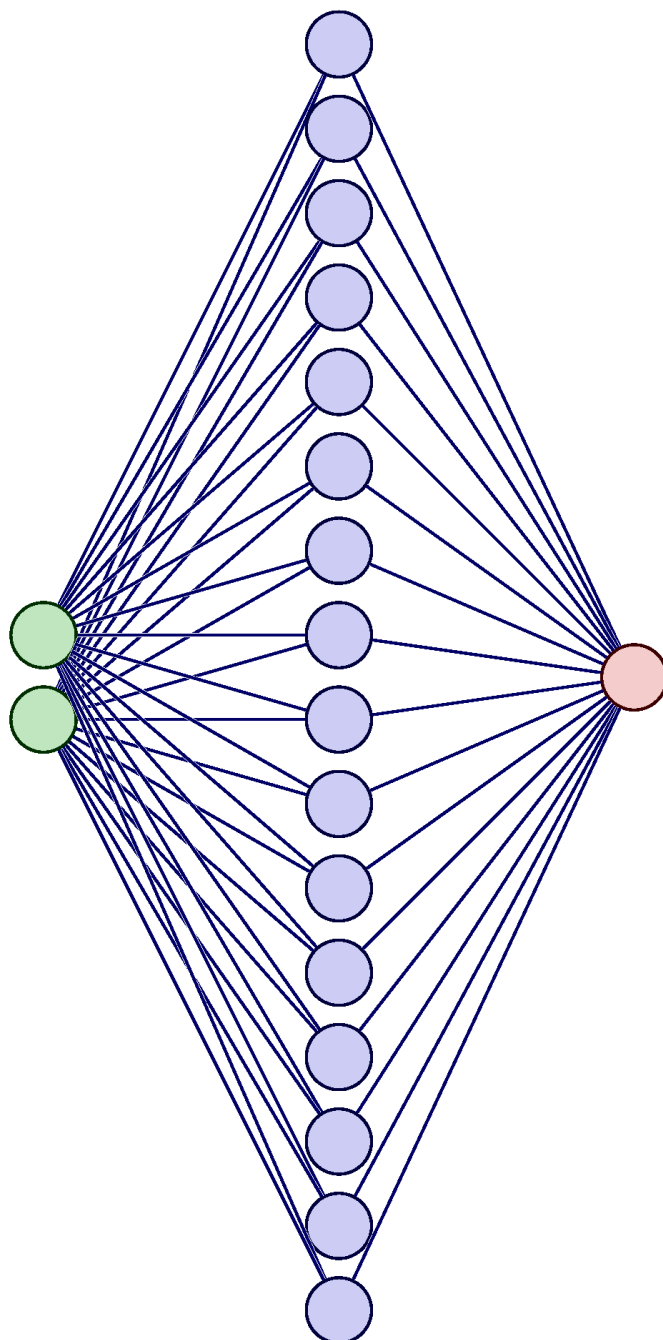


7.2 What does an Artificial Neural Network learn?



Exercise 8: An Artificial Neural Network

Label the graph:



Exercise 9:

How are you smarter than AI?